

DEEP-TECH LEAP

India's deep-tech ecosystem is at an inflection point, with start-ups moving beyond assembly to indigenous technologies, and confronting the harder challenge of scaling them globally

BY PALAK AGARWAL

THE NEXT GENERATION OF

Indian start-ups is not building another app. Instead, it's trying to build things India has traditionally imported—from microchips and rocket engines to advanced energy systems.

It's taking on problems that require years of research, specialised engineering, expensive prototypes and, often, technologies that have never before been commercially built in India.

For instance, AGNIT Semiconductors is building Gallium Nitride (GaN) semiconductors in India. GaN is a semiconductor that outperforms silicon. The company has filed for over 20 patents and has five products in field trials; two are completely indigenous.

Exponent Energy has filed for 37 patents and built a battery and charging technology designed to charge in 15 minutes. E-Plane is developing an electric aircraft in which more than 80% of roughly 1,500 components are of Indian origin.

Astrobase is working on an 800 kilo-Newton full-thrust combustion engine for a medium-lift rocket capable of carrying eight tonnes to

low orbit. D-Propulse is developing a rotating detonation engine, betting on a fundamentally different approach to propulsion, one of India's biggest aerospace and defence gaps, for future applications.

Together, they point to something bigger happening in India's entrepreneurial ecosystem—it is beginning to move from the application layer to the technology layer.

Greater localisation across the value chain, driven by strategic priorities, global competitiveness, policy reforms and a maturing talent pool are behind the shift, says P.S. Easwaran, Partner and leader of the supply chain practice at Deloitte India.

And the shift is beginning to show up in data.

INFLECTION POINT

According to The Kae Capital India Deeptech Report 2025, the deep-tech ecosystem in the country is at "inflection point", with more ventures being incubated, funded and reaching successful exits.

It has identified robotics and automation, energy tech, agritech, cybersecurity and consumer deeptech as areas with significant



potential and advanced materials and semiconductors as technologies that can underpin multiple industries. Spacetech and quantum technologies are areas in which India could potentially play a leading global role.

The report notes that around 20% of global chip-design talent resides in India, but fabrication, Outsourced Semiconductor Assembly And Test (OSAT) and packaging capabilities have been historically missing. That landscape is now changing, with the India Semiconductor Mission and large policy support pushing the